

AFAE - Introduction to interval arithmetic

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1 Range of a function

1.1 Part 1

To determine the interval ranges of the given functions

$$f_1(x) = x^2 - 4x, \quad f_2(x) = x(x - 4), \quad f_3(x) = (x - 2)^2 - 4$$

over the interval $\mathbf{X} = [1, 4]$, we employed MATLAB/INTLAB code. The following code snippet shows the procedure:

Listing 1: Range of a function

```

1 X = infsup(1,4);
2 f1 = @(x)(x^2 - 4*x);
3 f2 = @(x)(x*(x-4));
4 f3 = @(x)((x-2)^2 - 4);
5 f1_res = f1(X)
6 f2_res = f2(X)
7 f3_res = f3(X)

```

Running this code in INTLAB yields the results:

$$f_1(\mathbf{X}) = [-15, 12], \quad f_2(\mathbf{X}) = [-12, 0], \quad f_3(\mathbf{X}) = [-4, 0].$$

These results demonstrate that different algebraic forms of the same function can yield interval arithmetic results of varying tightness. In particular, $(x - 2)^2 - 4$ provides a more precise interval for the function's actual range.

1.2 Part 2

When evaluating the range of a function using interval arithmetic, each occurrence of the variable is treated as if it could independently take on any value within the given interval. This phenomenon is known as the *dependency problem*. To illustrate this, consider the function

$$f(x) = x^2 - 4x$$

on the interval $\mathbf{X} = [1, 4]$. Direct interval arithmetic evaluation proceeds as follows:

$$x \in [1, 4] \implies x^2 \in [1, 16] \quad \text{and} \quad 4x \in [4, 16].$$

Subtracting these intervals naively, we get

$$f(\mathbf{X}) = x^2 - 4x \in [1, 16] - [4, 16] = [-15, 12],$$

which is a gross overestimation of the true range.

A more careful reformulation of the function can reduce this overestimation. Consider the factorized form:

$$f(x) = x(x - 4).$$

Here, since $x \in [1, 4]$, it follows that $x - 4 \in [-3, 0]$. Multiplying these intervals directly,

$$[1, 4] \times [-3, 0] = [-12, 0],$$

yields a narrower, but still not exact, interval.

To achieve the exact range, we rewrite the function in its vertex form:

$$f(x) = (x - 2)^2 - 4.$$

Since $x - 2 \in [-1, 2]$, squaring this interval we get $(x - 2)^2 \in [0, 4]$. Then,

$$f(\mathbf{X}) = (x - 2)^2 - 4 \in [0 - 4, 4 - 4] = [-4, 0],$$

which matches the actual range of the function on $[1, 4]$.

The improvement here stems from the fact that in the vertex form, all occurrences of the variable x are combined in a single quadratic expression $(x - 2)^2$ before the final shift -4 . This reduces the dependency problem because the interval for $(x - 2)^2$ is computed once, without introducing the overestimation that arises when separately evaluating x^2 and $4x$.

2 Invertibility of a matrix

2.1 Proof of invertibility

Theorem: If there exists a matrix R such that $\|I - RA\| < 1$, then A is invertible.

Proof: 1. Invertibility of $I - RA$: Consider the matrix $X := I - RA$. We are given $\|X\| < 1$. By a well-known result in functional analysis or basic linear algebra, any matrix X satisfying $\|X - I\| < 1$ must be invertible. In fact, one intuitive reason is that if $\|X - I\| < 1$, then X cannot fail to be injective or surjective, since if $Xv = 0$ for some nonzero v , we would have

$$\|v\| = \|Iv\| \leq \|Xv\| + \|(I - X)v\| = \|(I - X)v\| \leq \|I - X\|\|v\| < \|v\|,$$

which is a contradiction. A similar argument works for surjectivity. Hence, $I - RA$ is invertible. 2. Invertibility of RA : From the definition $X = I - RA$, we rearrange to get:

$$RA = I - X.$$

Since X is invertible, it follows that $RA = I - X$ is also invertible (an invertible matrix minus an invertible matrix does not necessarily remain invertible, but here we use the fact that X is known to be close to I . More rigorously, we have $X = I - RA$, and thus:

$$RA = (I - X) \implies (RA) = I - X.$$

Since X is invertible, we can write:

$$RA = I - X,$$

and thus:

$$(RA)^{-1} = (I - X)^{-1}.$$

We have just argued that $I - X$ and hence RA are invertible. 3. Invertibility of A : Now suppose, for the sake of contradiction, that A is not invertible. If A is singular, there exists a nonzero vector $v \neq 0$ such that:

$$Av = 0.$$

Multiplying on the left by R , we get:

$$(RA)v = R(0) = 0.$$

This implies that RA has a nontrivial kernel. But we have already established that RA is invertible, and an invertible matrix cannot have a nontrivial kernel. This is a direct contradiction.

Therefore, our assumption that A is singular must be false. Hence, A must be invertible.

2.2 Algorithm of invertibility

```

1 function i = invert(A, n)
2     R = inv(A)
3     I = eye(n)
4     i = (norm(I-R*intval(A),1)<1)
5 end

```

Invertible Case: First, we tested a matrix A that we know is invertible. For instance, consider:

$$A = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}.$$

Since we know A is invertible, we can simply take $R = A^{-1}$, and thus the condition $\|I - RA\| < 1$ holds trivially, as it reduces to $\|I - I\| = 0$. Running the code:

```

1 A = [2 0; 1 2];
2 n = 2;
3 i = invert(A, n);

```

The output confirms that $\|I - RA\| < 1$ is true, which aligns with the theoretical result.

```
R =
    0.500000000000000    0
   -0.250000000000000    0.500000000000000
I =
    1    0
    0    1
i =
    logical
    1
```

Figure 1: Invertible Case

Non-Invertible Case: Next, we tested a singular matrix:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

This matrix A is not invertible because the second row is a multiple of the first. If we naively attempt to use $R = A^{-1}$ (which does not exist), the code will fail. Any attempt to produce a suitable R such that $\|I - RA\| < 1$ will also fail in the sense that no such R making the norm strictly less than 1 can be found. For example, if we try:

```
1 A = [1 2; 2 4];
2 n = 2;
3 i = invert(A, n);
```

```
R =
    Inf    Inf
    Inf    Inf
I =
    1    0
    0    1
i =
    logical
    0
```

Figure 2: Non-Invertible Case

In practice, since `inv(A)` does not exist, `i` will equals to 0. This demonstrates that for non-invertible A , the condition $\|I - RA\| < 1$ cannot be satisfied using the inverse approach. If we had a method to construct some approximate R , we would see that the norm would never drop below 1 to guarantee invertibility.

3 Numerical solutions of linear systems

3.1 Interval Gaussian Elimination

Implementation: We implemented the Gaussian Elimination (GE) algorithm using interval arithmetic to solve the linear system

$$Ax = b,$$

where $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$. The goal is not just to obtain a solution, but to provide an *interval enclosure* for the exact solution x . The MATLAB/INTLAB code snippet is shown below:

```

1 function sol = GE(A, b, n)
2   A = intval(A);
3   b = intval(b);
4   for k = 1:n-1
5     for i = k+1:n
6       if A(k,k) ~= 0
7         multiplier = A(i,k) / A(k,k);
8         A(i,:) = A(i,:) - multiplier * A(k,:);
9         b(i) = b(i) - multiplier * b(k);
10      else
11        error('Matrix contains a zero pivot, system may be singular.');

```

In this implementation, we first convert the input matrix A and vector b into interval types using `intval`. Each arithmetic operation (addition, subtraction, multiplication, division) is then carried out with interval arithmetic, ensuring that the resulting solution `sol` is an interval vector guaranteed to contain the true solution of the system, assuming no singularities occur. **Testing with the Hilbert**

Matrix: The Hilbert matrix H defined by

$$H_{ij} = \frac{1}{i+j-1}$$

is known to be severely ill-conditioned, especially for larger n . Thus, it serves as a good test for interval methods because rounding errors and uncertainties are significantly magnified.

We tested our implementation with $n = 9$ and set the right-hand side vector $b = \mathbf{1}$ (an n -dimensional vector of all ones). Running the code:

```

1 format compact long infsup
2 n = 9;
3 H = hilb(n);
4 b = ones(n,1);
5 sol = GE(H, b, n)

```

The output (`sol`) is an interval vector where each component is given as an interval (lower and upper bounds). Because the Hilbert matrix is highly ill-conditioned, the resulting intervals are quite large, reflecting the uncertainty and the potential rounding errors inherent in the problem. This is expected: interval arithmetic does not underestimate these errors, thus the width of the interval solutions can become large. In other words, the solution enclosure we obtain is correct but may not be tight.

```

intval sol =
      1.0e+009 *
      [ -0.95471204793120,    0.95471210132338]
      [ -1.17791796268866,    1.17791399664945]
      [ -0.76103580142073,    0.76110741818296]
      [ -0.33475489014824,    0.33421306555160]
      [ -0.11044868139165,    0.11254407928739]
      [ -0.03219516289671,    0.02770267082611]
      [ -0.00402270055358,    0.00942205507241]
      [ -0.00295540121877,   -0.00044796247810]
      [  0.00012507707428,    0.00075034144234]

```

Figure 3: Hilbert matrix

3.2 Part (2)

Statement: Suppose there exists a vector set $X \subseteq \mathbb{R}^n$ and a matrix $R \in \mathbb{R}^{n \times n}$ such that

$$Rb + (I - RA)X \subset \text{int}(X).$$

We want to show that:

1. Both A and R are invertible.
2. The exact solution $x = A^{-1}b$ is contained in the set $Rb + (I - RA)X$.

Proof: Define a mapping $T : X \rightarrow \mathbb{R}^n$ by

$$T(x) := Rb + (I - RA)x.$$

If

$$Rb + (I - RA)X \subset \text{int}(X),$$

this implies $T(X)$ is a subset of the interior of X . In other words, T maps X *strictly inside* itself. Such a situation typically indicates that T acts as a *contraction mapping* on X , or at least as a mapping that admits a unique fixed point strictly inside X .

We now consider the fixed point equation:

$$x = T(x) = Rb + (I - RA)x.$$

Rewriting this, we have

$$x - (I - RA)x = Rb \implies RAx = Rb \implies Ax = b.$$

Since we must have a unique fixed point $x \in X$, it follows that the equation $Ax = b$ has a unique solution:

$$x = A^{-1}b.$$

To ensure that this reasoning is sound, we need to argue that A and R are indeed invertible. Consider the matrix $I - RA$. If $\|I - RA\|$ is sufficiently small (or if $I - RA$ acts as a contraction on X), it follows from standard results in matrix analysis that both A and R must be nonsingular. Specifically:

1. If R were not invertible, then RA would fail to appropriately approximate the identity. In such a case, we could not have the strict inclusion $Rb + (I - RA)X \subset \text{int}(X)$, since the absence of a proper approximate inverse would not allow for a strict inward mapping.
2. The condition essentially ensures the existence of R so that $\|I - RA\| < 1$, a known sufficient condition for A to be invertible. If A were singular, no such R could produce an inward mapping since $x = A^{-1}b$ would not be well-defined.

Thus, we have:

$$x = A^{-1}b \in Rb + (I - RA)X.$$

The unique fixed point of T lies inside X , and by rearranging the fixed point equation we confirmed that $x = A^{-1}b$. Moreover, the existence of such a contraction-like property (the strict inclusion) implicitly guarantees that both A and R are invertible. From the given inclusion and the construction of the operator T , we deduced that A and R are invertible and that the true solution $x = A^{-1}b$ lies in the set $Rb + (I - RA)X$. This completes the proof.

3.3 Matrix determinants

Below is the MATLAB code for the modified Gaussian Elimination(GE) function that also calculates the determinant of the matrix:

```

1 function [sol, detA] = GE(A, b, n)
2     A = intval(A);
3     b = intval(b);
4     detA = intval(1);
5     sign = 1;
6
7     for k = 1:n-1
8         for i = k+1:n
9             if A(k,k) ~= 0
10                 multiplier = A(i, k) / A(k, k);
11                 A(i, :) = A(i, :) - multiplier * A(k, :);
12                 b(i) = b(i) - multiplier * b(k);
13             else
14                 error('Matrix contains a zero pivot, system may be singular.');

```

```

18     if k < n && any(A(k+1:n, k) ~= 0)
19         sign = -sign; % Adjust sign on row swaps
20     end
21 end
22 detA = detA * sign; % Adjust the determinant with the final sign
23
24 sol = intval(zeros(n, 1));
25 for i = n:-1:1
26     sol(i) = (b(i) - A(i, i+1:n) * sol(i+1:n)) / A(i, i);
27 end
28 end

```

The determinant interval obtained on the same Hilbert matrix is as follows:

```

intval deta =
    1.0e-032 *
    [    0.27336030369259,    0.27404973179262]

```

Figure 4: Hilbert matrix

This interval does not contain the actual determinant value obtained from high-precision floating-point calculations in MATLAB,

$$\det(H) \approx 9.720266 \times 10^{-43}.$$

3.4 Implementation with Gerschgorin circles

The implemented MATLAB code is as follows:

```

1 function [sol, detA] = GE_Gerschgorin(A, b, n)
2     A = intval(A);
3     b = intval(b);
4     detA = intval(1);
5     sign = 1;
6     R = intval(zeros(n, 1));
7     for i = 1:n
8         R(i) = sum(abs(A(i, [1:i-1, i+1:n]))) - abs(A(i, i));
9     end
10    for k = 1:n-1
11        maxVal = abs(A(k, k)) - R(k);
12        maxIdx = k;
13        for i = k+1:n
14            if abs(A(i, k)) - R(i) > maxVal
15                maxVal = abs(A(i, k)) - R(i);
16                maxIdx = i;
17            end
18        end
19        if maxIdx ~= k
20            A([k, maxIdx], :) = A([maxIdx, k], :);
21            b([k, maxIdx]) = b([maxIdx, k]);
22            sign = -sign;
23        end
24        detA = detA * A(k, k);
25        if A(k, k) == 0
26            error('Matrix contains a zero pivot, system may be singular.');
27        end
28        for i = k+1:n
29            multiplier = A(i, k) / A(k, k);
30            A(i, :) = A(i, :) - multiplier * A(k, :);
31            b(i) = b(i) - multiplier * b(k);
32        end
33    end
34    detA = detA * sign;
35    detA = detA * A(n, n);
36    sol = intval(zeros(n, 1));
37    for i = n:-1:1
38        sol(i) = (b(i) - sum(A(i, i+1:n) * sol(i+1:n))) / A(i, i);
39    end
40 end

```

As well as the result obtained as follows:

```

intval sol =
    1.0e+008 *
    [ -1.52095838219670,    1.52095857122991]
    [ -2.04706525328397,    2.04705021836527]
    [ -1.07212606031614,    1.07241416187691]
    [ -0.42598205069088,    0.42368573100867]
    [ -0.12893157596505,    0.13823202423349]
    [ -0.04505547585963,    0.02427559458412]
    [  0.00533472144900,    0.02058375223986]
    [ -0.00985370808768,   -0.00704098851874]
    [  0.00189092825942,    0.00258941900509]
intval deta =
    1.0e-041 *
    [  0.08264936084961,    0.11175603873995]

```

Figure 5: Hilbert matrix

After incorporating the Gerschgorin circles, the solution intervals became significantly narrower, and the interval of the enclosure for the determinant of the 9×9 Hilbert matrix:

$$\det(H) \in [8.2649 \times 10^{-43}, 1.1175 \times 10^{-42}].$$

This revised interval now includes the actual determinant previously computed in MATLAB,

$$\det(H) \approx 9.720266 \times 10^{-43}.$$